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SECR1033 - Computer Organization and Architecture

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Question 1

$$a) \quad 23 = 0001 \ 0111_2$$

$$62 = 0011 \ 1110$$

$$1110 \ 1000 \text{ — 1's complement}$$

$$+ \quad \quad \quad 1 \text{ — 2's complement}$$

$$\hline 1110 \ 1001$$

$$+ \quad 1110 \ 1001$$

$$+ \quad 1100 \ 0010$$

$$\hline 1010 \ 1011$$

ignored

$$1100 \ 0001 \text{ — 1's complement}$$

$$+ \quad \quad \quad 1 \text{ — 2's complement}$$

$$\hline 1100 \ 0010$$

$$+ \quad 0101 \ 0100 \text{ — 1's complement}$$

$$+ \quad \quad \quad 1 \text{ — 2's complement}$$

$$\hline 0101 \ 0101$$

$$0101 \ 0101_2 = 85_{10}$$

∴ as the MSB is 1, hence the number is -85

No overflow occurred

b) $21_{10} \times -14_{10}$

$-21_{10} \times 14_{10}$

MC MP

$21_{10} = 0000\ 0001\ 0101$

$1111\ 1110\ 1010$ — 1's complement

+ 1 — 2's complement

$1111\ 1110\ 1011$

iteration	steps	MP	MC	Product
0	initial value	001110	1111 1110 1011	0000 0000 0000
1	No Operation shift MC to the left shift MP to the right	000 111	1111 1101 0110	-
2	$P = P + MC$ shift MC to the left shift MP to the right	000 011	1111 1010 1100	1111 1101 0110
3	$P = P + MC$ shift MC to the left shift MP to the right	000 001	1111 0101 1000	1111 1000 0010
4	$P = P + MC$ shift MC to the left shift MP to the right	000 000	1110 1011 0000	1110 1101 1010
5	no operation shift MC to the left shift MP to the right	000 000	1101 0110 0000	-
6	no operation shift MC to the left shift MP to the right	000 000	1010 1100 0000	-

b) 0001 0010 0101 — 1's complement
+ 1 — 2's complement

0001 0010 0110

0001 0010 0110₂ = 294

∴ since the MSB is 1, the number will be -294

Question 2

a) $M = 1.1101_2 \times 2^6$
 $N = -1.01_2 \times 2^{-10}$

i) $(6) + (-10) = -4$
 $(-4) + 127 = 123$

ii)

$$\begin{array}{r}
 1.1101 \\
 \times 1.01 \\
 \hline
 11101 \\
 00000 \\
 11101 \\
 \hline
 10.01001
 \end{array}
 \rightarrow 10.01001 \times 2^{-4}$$

iii) 1.0010001×2^{-3}

iv) -1.0010001×2^{-3}

b) for single precision format; Bias value = 127

$$\begin{aligned}
 E^B &= (-3) + 127 \\
 &= 124_{10} \\
 &= 01111100
 \end{aligned}$$

1 bit	8 bits	23 bits
1	01111100	0010 0010 0000 0000 0000 0000
sign	Bias Exponent	Fraction

Question 3

(a) $e' = 2$

$$E_b = 16 + 2$$

$$= 18$$

$18_{10} = 10010_2$

0.1110111×2^2 (normalized)

1bit	5bit	8bit
0	10010	1110 1110

$$\begin{array}{r} 2 \overline{) 13} \\ 2 \overline{) 6} - 1 \\ 2 \overline{) 3} - 0 \\ 2 \overline{) 1} - 1 \\ 0 - 1 \end{array}$$

$0.625 \times 2 = 1.25$

$0.25 \times 2 = 0.50$

$= 0.10$

$= 1101$

$-13.625_{10} = -1101.10_2$

$= -0.110110 \times 2^4$

$b = 2^{5-1}$

$= 16$

$E_b = 16 + 4$

$= 20$

$= 10100$

1	10100	1101 1000
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$$(C) C = 1.1011 \times 2^{-3}$$

$$D = -1.0001 \times 2^{-4}$$

$$= -1.0001 \times 2^{-4} \times 2^1$$

$$= 0 - 0.10001 \times 2^{-3}$$

$$1.10110 \times 2^{-3}$$

$$+ - 0.10001 \times 2^{-3}$$

$$\begin{array}{r} 1.10110 \\ - 0.10001 \\ \hline 1.00101 \end{array} \times 2^{-3}$$

(4 binary digits)

$$= -1.0000 \times 2^{-3}$$

(d)	2 2	0.225 × 2 = 0.45
	2 1 - 0	0.45 × 2 = 0.90
	0 - 1	0.90 × 2 = 1.80
	= 10	0.80 × 2 = 1.60
		= 0.0011

$$2.225 \times 10^1 = 10.0011 \times 2^1$$

$$= 1.00011 \times 2^2$$

$$1.234 \times 10^0$$

2 1	0.234 × 2 = 0.468
0 - 1	0.468 × 2 = 0.936
= 1	0.936 × 2 = 1.872
	0.872 × 2 = 1.744

$$1.234 \times 10^0 = 1.0011 \times 2^0$$

$$2 + 6 = 2$$

If bias are consider

$$127 + 2 = 129$$

$$\begin{array}{r} 1.00011 \\ \times 1.00110 \\ \hline 000000 \\ 100011 \\ 100011 \\ 000000 \\ 000000 \\ 100011 \\ \hline 1.0100110010 \times 2^2 \end{array}$$

$$= 1.0100 \times 2^2 \text{ (4 significant bit)}$$